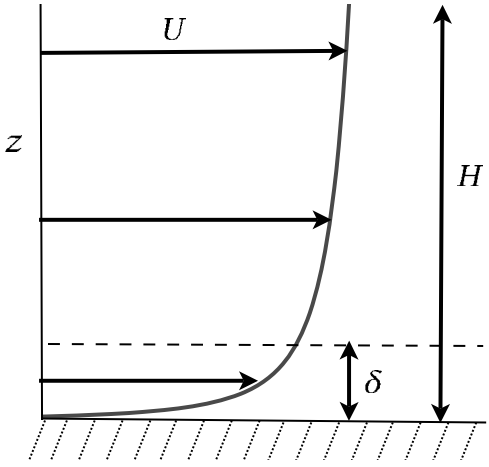


Fig. 2.10 An idealized boundary layer. The values of a field, such as velocity, U, may vary rapidly in a boundary in order to satisfy the boundary conditions at a rigid surface. The parameter δ is a measure of the boundary layer thickness, and H is a typical scale of variation away from the boundary.

2.12 THE EKMAN LAYER

In the final topic of this chapter, we return to geostrophic flow and consider the effects of friction. The fluid fields in the interior of a domain are often set by different physical processes than those occurring at a boundary, and consequently often change rapidly in a thin *boundary layer,* as in Fig. 2.10. Such boundary layers nearly always involve one or both of viscosity and diffusion, because these appear in the terms of highest differential order in the equations of motion, and so are responsible for the number and type of boundary conditions that the equations must satisfy — for example, the presence of molecular viscosity leads to the condition that the tangential flow (as well as the normal flow) must vanish at a rigid surface.

In many boundary layers in non-rotating flow the dominant balance in the momentum equation is between the advective and viscous terms. In some contrast, in large-scale atmospheric and oceanic flow the effects of rotation are large, and this results in a boundary layer, known as the *Ekman layer,* in which the dominant balance is between Coriolis and frictional or stress terms.[12] Now, the direct effects of molecular viscosity and diffusion are nearly always negligible at distances more than a few millimetres away from a solid boundary, but it is inconceivable that the entire boundary layer between the free atmosphere (or free ocean) and the surface is only a few millimetres thick. Rather, in practice a balance occurs between the Coriolis terms and the forces due to the stress generated by small-scale turbulent motion, and this gives rise to a boundary layer that has a typical depth of a few tens to several hundreds of metres. Because the stress arises from the turbulence we cannot with confidence determine its precise form; thus, we should try to determine what general properties Ekman layers may have that are *independent* of the precise form of the friction.

The atmospheric Ekman layer occurs near the ground, and the stress at the ground itself is due to the surface wind (and its vertical variation). In the ocean the main Ekman layer is near the surface, and the stress at ocean surface is largely due to the presence of the overlying wind. There is also a weak Ekman layer at the bottom of the ocean, analogous to the atmospheric Ekman layer. To analyse all these layers, let us assume the following.

- ⋆ The Ekman layer is Boussinesq. This is a very good assumption for the ocean, and a reasonable one for the atmosphere if the boundary layer is not too deep.

- ⋆ The Ekman layer has a finite depth that is less than the total depth of the fluid, this depth being given by the level at which the frictional stresses essentially vanish. Within the Ekman layer, frictional terms are important, whereas geostrophic balance holds beyond it.
- ⋆ The nonlinear and time-dependent terms in the equations of motion are negligible, hydrostatic balance holds in the vertical, and buoyancy is constant, not varying in the horizontal.
- ⋆ As needed, we shall assume that friction can be parameterized by a viscous term of the form $\rho_0^{-1}\partial\boldsymbol{\tau}/\partial z = A\,\partial^2\boldsymbol{u}/\partial z^2$, where A is constant and $\boldsymbol{\tau}$ is the stress. [In general, stress is a tensor, τ_{ij}, with an associated force given by $F_i = \partial\tau_{ij}/\partial x_j$, summing over the repeated index. It is common in geophysical fluid dynamics that the vertical derivative dominates, and in this case the force is $\boldsymbol{F} = \partial\boldsymbol{\tau}/\partial z$. We still use the word stress for $\boldsymbol{\tau}$, but it now refers to a vector whose derivative in a particular direction (z in this case) is the force on a fluid.] In laboratory settings A may be the molecular viscosity, whereas in the atmosphere and ocean it is a so-called *eddy viscosity*. (In turbulent flows momentum is transferred by the near-random motion of small parcels of fluid and, by analogy with the motion of molecules that produces a molecular viscosity, the associated stress is approximately represented, or parameterized, using a turbulent or eddy viscosity that may be orders of magnitude larger than the molecular one.)

2.12.1 Equations of motion and scaling

Frictional–geostrophic balance in the horizontal momentum equation is:

$$\boldsymbol{f}\times\boldsymbol{u} = -\nabla_z\phi + \frac{\partial\tilde{\boldsymbol{\tau}}}{\partial z}. \tag{2.273}$$

where $\tilde{\boldsymbol{\tau}} \equiv \boldsymbol{\tau}/\rho_0$ is the kinematic stress and $\boldsymbol{f} = f\mathbf{k}$, where the Coriolis parameter f is allowed to vary with latitude. If we model the stress with an eddy viscosity, (2.273) becomes

$$\boldsymbol{f}\times\boldsymbol{u} = -\nabla_z\phi + A\frac{\partial^2\boldsymbol{u}}{\partial z^2}. \tag{2.274}$$

The vertical momentum equation is $\partial\phi/\partial z = b$, i.e., hydrostatic balance, and, because buoyancy is constant, we may without loss of generality write this as

$$\frac{\partial\phi}{\partial z} = 0. \tag{2.275}$$

The equation set is completed by the mass continuity equation, $\nabla\cdot\boldsymbol{v} = 0$.

The Ekman number

We non-dimensionalize the equations by setting

$$(u,v) = U(\hat{u},\hat{v}), \quad (x,y) = L(\hat{x},\hat{y}), \quad f = f_0\hat{f}, \quad z = H\hat{z}, \quad \phi = \Phi\hat{\phi}, \tag{2.276}$$

where hatted variables are non-dimensional. H is a scaling for the height, and at this stage we will suppose it to be some height scale in the free atmosphere or ocean, not the height of

the Ekman layer itself. Geostrophic balance suggests that $\Phi = f_0 UL$. Substituting (2.276) into (2.274) we obtain

$$\hat{\boldsymbol{f}} \times \hat{\boldsymbol{u}} = -\hat{\nabla}\hat{\phi} + Ek\frac{\partial^2 \hat{\boldsymbol{u}}}{\partial \hat{z}^2}, \tag{2.277}$$

where the parameter

$$\boxed{Ek \equiv \left(\frac{A}{f_0 H^2}\right)}, \tag{2.278}$$

is the *Ekman number*, and it determines the importance of frictional terms in the horizontal momentum equation. If $Ek \ll 1$ then the friction is small in the flow interior where $\hat{z} = \mathcal{O}(1)$. However, the friction term cannot necessarily be neglected in the boundary layer because it is of the highest differential order in the equation, and so determines the boundary conditions; if Ek is small the vertical scales become small and the second term on the right-hand side of (2.277) remains finite. The case when this term is simply omitted from the equation is therefore a *singular limit,* meaning that it differs from the case with $Ek \to 0$. If $Ek \geq 1$ friction is important everywhere, but it is usually the case that Ek is small for atmospheric and oceanic large-scale flow, and the interior flow is very nearly geostrophic. (In part this is because A itself is only large near a rigid surface where the presence of a shear creates turbulence and a significant eddy viscosity.)

Momentum balance in the Ekman layer

For definiteness, suppose the fluid lies above a rigid surface at $z = 0$. Sufficiently far away from the boundary the velocity field is known, and we suppose this flow to be in geostrophic balance. We then write the velocity field and the pressure field as the sum of the interior geostrophic part, plus a boundary layer correction:

$$\hat{\boldsymbol{u}} = \hat{\boldsymbol{u}}_g + \hat{\boldsymbol{u}}_E, \qquad \hat{\phi} = \hat{\phi}_g + \hat{\phi}_E, \tag{2.279}$$

where the Ekman layer corrections, denoted with a subscript E, are negligible away from the boundary layer. Now, in the fluid interior we have, by hydrostatic balance, $\partial\hat{\phi}_g/\partial\hat{z} = 0$. In the boundary layer we still have $\partial\hat{\phi}_g/\partial\hat{z} = 0$ so that, to satisfy hydrostasy, $\partial\hat{\phi}_E/\partial\hat{z} = 0$. But because $\hat{\phi}_E$ vanishes away from the boundary we have $\hat{\phi}_E = 0$ everywhere. Thus, *there is no boundary layer in the pressure field.* Note that this is a much stronger result than saying that pressure is continuous, which is nearly always true in fluids; rather, it is a special result for Ekman layers.

Using (2.279) with $\hat{\phi}_E = 0$, the dimensional horizontal momentum equation (2.273) becomes, in the Ekman layer,

$$\boldsymbol{f} \times \boldsymbol{u}_E = \frac{\partial \tilde{\boldsymbol{\tau}}}{\partial z}. \tag{2.280}$$

The dominant force balance in the Ekman layer is thus between the Coriolis force and the friction. We can determine the thickness of the Ekman layer if we model the stress with an eddy viscosity so that

$$\boldsymbol{f} \times \boldsymbol{u}_E = A\frac{\partial^2 \boldsymbol{u}_E}{\partial z^2}, \tag{2.281}$$

or, non-dimensionally,

$$\hat{\boldsymbol{f}} \times \hat{\boldsymbol{u}}_E = Ek\frac{\partial^2 \hat{\boldsymbol{u}}_E}{\partial \hat{z}^2}. \tag{2.282}$$

It is evident this equation can only be satisfied if $\hat{z} \neq \mathcal{O}(1)$, implying that H is not a proper scaling for z in the boundary layer. Rather, if the vertical scale in the Ekman layer is $\hat{\delta}$ (meaning $\hat{z} \sim \hat{\delta}$) we must have $\hat{\delta} \sim Ek^{1/2}$. In dimensional terms this means the thickness of the Ekman layer is

$$\delta = H\hat{\delta} = HEk^{1/2} \tag{2.283}$$

or

$$\boxed{\delta = \left(\frac{A}{f_0}\right)^{1/2}}. \tag{2.284}$$

[This estimate also emerges directly from (2.281).] Note that (2.283) can be written as

$$Ek = \left(\frac{\delta}{H}\right)^2. \tag{2.285}$$

That is, the Ekman number is equal to the square of the ratio of the depth of the Ekman layer to an interior depth scale of the fluid motion. In laboratory flows where A is the molecular viscosity we can thus estimate the Ekman layer thickness, and if we know the eddy viscosity of the ocean or atmosphere we can estimate their respective Ekman layer thicknesses. We can invert this argument and obtain an estimate of A if we know the Ekman layer depth. In the atmosphere, deviations from geostrophic balance are very small in the atmosphere above 1 km, and using this gives $A \approx 10^2\,\mathrm{m^2\,s^{-1}}$. In the ocean Ekman depths are often 50 m or less, and eddy viscosities are about $0.1\,\mathrm{m^2\,s^{-1}}$.

2.12.2 Integral properties of the Ekman layer

What can we deduce about the Ekman layer without specifying the detailed form of the frictional term? Using dimensional notation we recall frictional–geostrophic balance,

$$\boldsymbol{f} \times \boldsymbol{u} = -\nabla\phi + \frac{1}{\rho_0}\frac{\partial \boldsymbol{\tau}}{\partial z}, \tag{2.286}$$

where $\boldsymbol{\tau}$ is zero at the edge of the Ekman layer. In the Ekman layer itself we have

$$\boldsymbol{f} \times \boldsymbol{u}_E = \frac{1}{\rho_0}\frac{\partial \boldsymbol{\tau}}{\partial z}. \tag{2.287}$$

Consider either a top or bottom Ekman layer, and integrate over its thickness. From (2.287) we obtain

$$\boldsymbol{f} \times \boldsymbol{M}_E = \boldsymbol{\tau}_T - \boldsymbol{\tau}_B, \tag{2.288}$$

where

$$\boldsymbol{M}_E = \int_{Ek} \rho_0 \boldsymbol{u}_E \,\mathrm{d}z \tag{2.289}$$

is the ageostrophic mass transport in the Ekman layer, and $\boldsymbol{\tau}_T$ and $\boldsymbol{\tau}_B$ are the respective stresses at the top and the bottom of the Ekman layer at hand. The stress at the top (bottom) will be zero in a bottom (top) Ekman layer and therefore, from (2.288),

$$\boxed{\begin{array}{ll} \text{top Ekman layer:} & \boldsymbol{M}_E = -\dfrac{1}{f}\mathbf{k} \times \boldsymbol{\tau}_T \\ \text{bottom Ekman layer:} & \boldsymbol{M}_E = \dfrac{1}{f}\mathbf{k} \times \boldsymbol{\tau}_B \end{array}}. \tag{2.290a,b}$$

The transport is thus at right angles to the stress at the surface, and proportional to the magnitude of the stress. These properties have a simple physical explanation: integrated over the depth of the Ekman layer the surface stress must be balanced by the Coriolis force, which in turn acts at right angles to the mass transport. A consequence of (2.290) is that the mass transports in adjacent oceanic and atmospheric Ekman layers are equal and opposite, because the stress is continuous across the ocean–atmosphere interface. Equation (2.290a) is particularly useful in the ocean, where the stress at the surface is primarily due to the wind, and is largely independent of the interior oceanic flow. In the atmosphere, the surface stress mainly arises as a result of the interior atmospheric flow, and to calculate it we need to parameterize the stress in terms of the flow.

Finally, we obtain an expression for the vertical velocity induced by an Ekman layer. The mass conservation equation is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{2.291}$$

Integrating this over an Ekman layer gives

$$\frac{1}{\rho_0} \nabla \cdot \boldsymbol{M}_{To} = -(w_T - w_B), \tag{2.292}$$

where $\boldsymbol{M}_{To}$ is the total (Ekman plus geostrophic) mass transport in the Ekman layer,

$$\boldsymbol{M}_{To} = \int_{Ek} \rho_0 \boldsymbol{u} \, \mathrm{d}z = \int_{Ek} \rho_0 (\boldsymbol{u}_g + \boldsymbol{u}_E) \, \mathrm{d}z \equiv \boldsymbol{M}_g + \boldsymbol{M}_E, \tag{2.293}$$

and w_T and w_B are the vertical velocities at the top and bottom of the Ekman layer; the former (latter) is zero in a top (bottom) Ekman layer. Equations (2.293) and (2.288) give

$$\mathbf{k} \times (\boldsymbol{M}_{To} - \boldsymbol{M}_g) = \frac{1}{f} (\boldsymbol{\tau}_T - \boldsymbol{\tau}_B). \tag{2.294}$$

Taking the curl of this (i.e., cross-differentiating) gives

$$\nabla \cdot (\boldsymbol{M}_{To} - \boldsymbol{M}_g) = \mathrm{curl}_z [(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B)/f], \tag{2.295}$$

where the curl_z operator on a vector $\boldsymbol{A}$ is defined by $\mathrm{curl}_z \boldsymbol{A} \equiv \partial_x A_y - \partial_y A_x$. Using (2.292) we obtain, for top and bottom Ekman layers respectively,

$$\boxed{w_B = \frac{1}{\rho_0} \left(\mathrm{curl}_z \frac{\boldsymbol{\tau}_T}{f} + \nabla \cdot \boldsymbol{M}_g \right), \qquad w_T = \frac{1}{\rho_0} \left(\mathrm{curl}_z \frac{\boldsymbol{\tau}_B}{f} - \nabla \cdot \boldsymbol{M}_g \right)} , \tag{2.296a,b}$$

where $\nabla \cdot \boldsymbol{M}_g = -(\beta/f) \boldsymbol{M}_g \cdot \mathbf{j}$ is the divergence of the geostrophic transport in the Ekman layer, and this is often small compared to the other terms in these equations. Thus, friction induces a vertical velocity at the edge of the Ekman layer, proportional to the curl of the stress at the surface, and this is perhaps the most used result in Ekman layer theory. Numerical models sometimes do not have the vertical resolution to explicitly resolve an Ekman layer, and (2.296) provides a means of *parameterizing* the Ekman layer in terms of resolved or known fields. It is particularly useful for the top Ekman layer in the ocean, where the stress can be regarded as a given function of the overlying wind.

2.12.3 Explicit solutions. I: a bottom boundary layer

We now assume that the frictional terms can be parameterized as an eddy viscosity and calculate the explicit form of the solution in the boundary layer. The frictional–geostrophic balance may be written as

$$\boldsymbol{f} \times (\boldsymbol{u} - \boldsymbol{u}_g) = A \frac{\partial^2 \boldsymbol{u}}{\partial z^2}, \tag{2.297a}$$

where

$$f(u_g, v_g) = \left(-\frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial x} \right). \tag{2.297b}$$

We continue to assume there are no horizontal gradients of temperature, so that, via thermal wind, $\partial u_g / \partial z = \partial v_g / \partial z = 0$.

Boundary conditions and solution

Appropriate boundary conditions for a bottom Ekman layer are

$$\text{at } z = 0: \qquad u = 0, \quad v = 0 \qquad \text{(the no slip condition)} \tag{2.298a}$$

$$\text{as } z \to \infty: \qquad u = u_g, \quad v = v_g \qquad \text{(a geostrophic interior).} \tag{2.298b}$$

Let us seek solutions to (2.297a) of the form

$$u = u_g + A_0 \mathrm{e}^{\alpha z}, \qquad v = v_g + B_0 \mathrm{e}^{\alpha z}, \tag{2.299}$$

where A_0 and B_0 are constants. Substituting into (2.297a) gives two homogeneous algebraic equations

$$A_0 f - B_0 A \alpha^2 = 0, \qquad -A_0 A \alpha^2 - B_0 f = 0. \tag{2.300a,b}$$

For non-trivial solutions the solvability condition $\alpha^4 = -f^2/A^2$ must hold, from which we find $\alpha = \pm(1 \pm \mathrm{i})\sqrt{f/2A}$. Using the boundary conditions we then obtain the solution

$$u = u_g - \mathrm{e}^{-z/d} \left[u_g \cos(z/d) + v_g \sin(z/d) \right] \tag{2.301a}$$

$$v = v_g + \mathrm{e}^{-z/d} \left[u_g \sin(z/d) - v_g \cos(z/d) \right], \tag{2.301b}$$

where $d = \sqrt{2A/f}$ is, within a constant factor, the depth of the Ekman layer obtained from scaling considerations. The solution decays exponentially from the surface with this e-folding scale, so that d is a good measure of the Ekman layer thickness. Note that the boundary layer correction depends on the interior flow, since the boundary layer serves to bring the flow to zero at the surface.

To illustrate the solution, suppose that the pressure force is directed in the y-direction (northwards), so that the geostrophic current is eastwards. Then the solution, the now-famous *Ekman spiral*, is plotted in Figs. 2.11 and 2.12. The wind falls to zero at the surface, and its direction just above the surface is northeastwards; that is, it is rotated by 45° to the left of its direction in the free atmosphere. Although this result is independent of the value of the frictional coefficients, it is dependent on the form of the friction chosen. The force balance in the Ekman layer is between the Coriolis force, the stress, and the pressure force. At the surface the Coriolis force is zero, and the balance is entirely between the northward pressure force and the southward stress force.

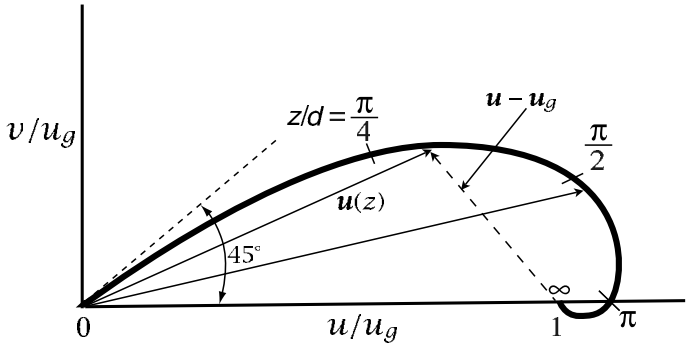


Fig. 2.11 The idealized Ekman layer solution in the lower atmosphere, plotted as a hodograph of the wind components: the arrows show the velocity vectors at a particular heights, and the curve traces out the continuous variation of the velocity. The values on the curve are of the non-dimensional variable z/d, where $d = (2A/f)^{1/2}$, and v_g is chosen to be zero.

Transport, force balance and vertical velocity

The cross-isobaric flow is given by (for $v_g = 0$)

$$V = \int_0^\infty v \, \mathrm{d}z = \int_0^\infty u_g \mathrm{e}^{-z/d} \sin(z/d) \, \mathrm{d}z = \frac{u_g d}{2}. \tag{2.302}$$

For positive f, this is to the left of the geostrophic flow — that is, down the pressure gradient. In the general case ($v_g \neq 0$) we obtain

$$V = \int_0^\infty (v - v_g) \, \mathrm{d}z = \frac{d}{2}(u_g - v_g). \tag{2.303}$$

Similarly, the additional zonal transport produced by frictional effects are, for $v_g = 0$,

$$U = \int_0^\infty (u - u_g) \, \mathrm{d}z = -\int_0^\infty \mathrm{e}^{-z/d} \sin(z/d) \, \mathrm{d}z = -\frac{u_g d}{2}, \tag{2.304}$$

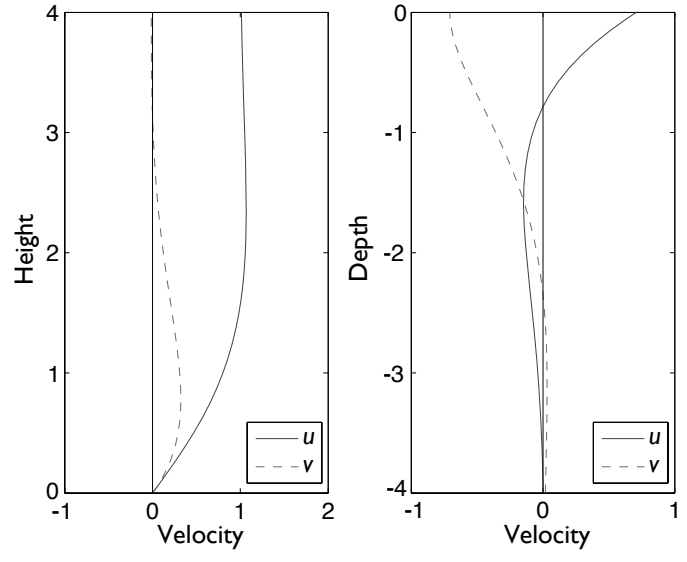


Fig. 2.12 Solutions for a bottom Ekman layer with a given flow in the fluid interior (left), and for a top Ekman layer with a given surface stress (right), both with $d = 1$. On the left we have $u_g = 1$, $v_g = 0$. On the right we have $u_g = v_g = 0$, $\tilde{\tau}_y = 0$ and $\sqrt{2}\tilde{\tau}_x/(fd) = 1$.

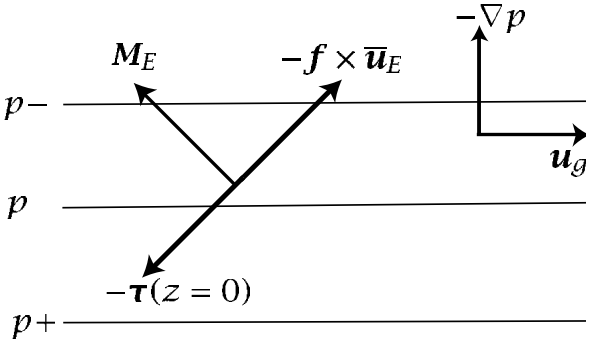


Fig. 2.13 A bottom Ekman layer, generated from an eastward geostrophic flow above it. An overbar denotes a vertical integral over the Ekman layer, so that $-\boldsymbol{f}\times\overline{\boldsymbol{u}}_E$ is the Coriolis force on the vertically integrated Ekman velocity. $\boldsymbol{M}_E$ is the frictionally induced boundary layer transport, and $\boldsymbol{\tau}$ is the stress.

and in the general case

$$U = \int_0^\infty (u - u_g)\,\mathrm{d}z = -\frac{d}{2}(u_g + v_g). \tag{2.305}$$

Thus, the total transport caused by frictional forces is

$$\boldsymbol{M}_E = \frac{\rho_0 d}{2}\left[-\mathbf{i}(u_g + v_g) + \mathbf{j}(u_g - v_g)\right]. \tag{2.306}$$

The total stress at the bottom surface $z = 0$ induced by frictional forces is

$$\tilde{\boldsymbol{\tau}}_B = A\frac{\partial \boldsymbol{u}}{\partial z}\bigg|_{z=0} = \frac{A}{d}\left[\mathbf{i}(u_g - v_g) + \mathbf{j}(u_g + v_g)\right], \tag{2.307}$$

using the solution (2.301). Thus, using (2.306), (2.307) and $d^2 = 2A/f$, we see that the total frictionally induced transport in the Ekman layer is related to the stress at the surface by $\boldsymbol{M}_E = (\mathbf{k}\times\boldsymbol{\tau}_B)/f$, reprising the result of the previous more general analysis, (2.296). From (2.307), the stress is at an angle of 45° to the left of the velocity at the surface. (However, this result is not generally true for all forms of stress.) These properties are illustrated in Fig. 2.13.

The vertical velocity at the top of the Ekman layer, w_E, is obtained using (2.306) or (2.307). If f if constant we obtain

$$w_E = -\frac{1}{\rho_0}\nabla\cdot\boldsymbol{M}_E = \frac{1}{f_0}\mathrm{curl}_z\tilde{\boldsymbol{\tau}}_B = \frac{d}{2}\zeta_g, \tag{2.308}$$

where ζ_g is the vorticity of the geostrophic flow. Thus, the vertical velocity at the top of the Ekman layer, which arises because of the frictionally-induced divergence of the cross-isobaric flow in the Ekman layer, is proportional to the geostrophic vorticity in the free fluid and is proportional to the Ekman layer height $\sqrt{2A/f_0}$.

Another bottom boundary condition

In the analysis above we assumed a *no slip* condition at the surface, namely that the velocity tangential to the surface vanishes. This is certainly appropriate if A is a molecular velocity, but in a turbulent flow, where A is interpreted as an eddy viscosity, the flow very close to the surface may be far from zero. Then, unless we wish to explicitly calculate the flow in an additional very thin viscous boundary layer the no-slip condition may be inappropriate. An

alternative, slightly more general boundary condition is to suppose that the stress at the surface is given by

$$\boldsymbol{\tau} = \rho_0 C \boldsymbol{u}, \tag{2.309}$$

where C is a constant. The surface boundary condition is then

$$A\frac{\partial \boldsymbol{u}}{\partial z} = C\boldsymbol{u}. \tag{2.310}$$

If C is infinite we recover the no-slip condition. If $C = 0$, we have a condition of no stress at the surface, also known as a *free slip* condition. For intermediate values of C the boundary condition is known as a 'mixed condition'. Evaluating the solution in these cases is left as an exercise for the reader (problem 2.28).

2.12.4 Explicit solutions. II: the upper ocean

Boundary conditions and solution

The wind provides a stress on the upper ocean, and the Ekman layer serves to communicate this to the oceanic interior. Appropriate boundary conditions are thus:

$$\text{at } z = 0: \qquad A\frac{\partial u}{\partial z} = \tilde{\tau}^x, \quad A\frac{\partial v}{\partial z} = \tilde{\tau}^y \qquad \text{(a given surface stress)} \tag{2.311a}$$

$$\text{as } z \to -\infty: \qquad u = u_g, \quad v = v_g \qquad \text{(a geostrophic interior)} \tag{2.311b}$$

where $\tilde{\boldsymbol{\tau}}$ is the given (kinematic) wind stress at the surface. Solutions to (2.297a) with (2.311) are found by the same methods as before, and are

$$u = u_g + \frac{\sqrt{2}}{fd}\mathrm{e}^{z/d}\left[\tilde{\tau}^x \cos(z/d - \pi/4) - \tilde{\tau}^y \sin(z/d - \pi/4)\right], \tag{2.312}$$

and

$$v = v_g + \frac{\sqrt{2}}{fd}\mathrm{e}^{z/d}\left[\tilde{\tau}^x \sin(z/d - \pi/4) + \tilde{\tau}^y \cos(z/d - \pi/4)\right]. \tag{2.313}$$

Note that the boundary layer correction depends only on the imposed surface stress, and not the interior flow itself. This is a consequence of the type of boundary conditions chosen, for in the absence of an imposed stress the boundary layer correction is zero — the interior flow already satisfies the gradient boundary condition at the top surface. Similar to the bottom boundary layer, the velocity vectors of the solution trace a diminishing spiral as they descend into the interior (Fig. 2.14, which is drawn for the Southern Hemisphere).

Transport, surface flow and vertical velocity

The transport induced by the surface stress is obtained by integrating (2.312) and (2.313) from the surface to $-\infty$. We explicitly find

$$U = \int_{-\infty}^{0} (u - u_g)\,\mathrm{d}z = \frac{\tilde{\tau}^y}{f}, \qquad V = \int_{-\infty}^{0} (v - v_g)\,\mathrm{d}z = -\frac{\tilde{\tau}^x}{f}, \tag{2.314}$$

which indicates that the ageostrophic transport is perpendicular to the wind stress, as noted previously from more general considerations. Suppose that the surface wind is eastward; in this case $\tilde{\tau}^y = 0$ and the solutions immediately give

$$u(0) - u_g = \tilde{\tau}^x/fd, \qquad v(0) - v_g = -\tilde{\tau}^x/fd. \tag{2.315}$$

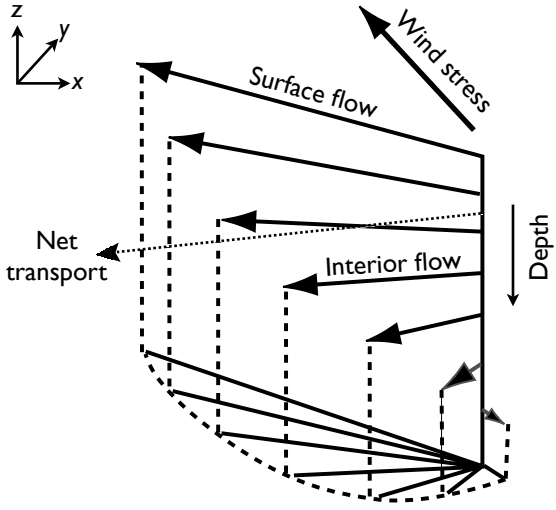


Fig. 2.14 An idealized Ekman spiral in a southern hemisphere ocean, driven by an imposed wind stress. A northern hemisphere spiral would be the reflection of this about the vertical axis. Such a clean spiral is rarely observed in the real ocean. The net transport is at right angles to the wind, independent of the detailed form of the friction. The angle of the surface flow is 45° to the wind only for a Newtonian viscosity.

Therefore the magnitudes of the frictional flow in the x and y directions are equal to each other, and the ageostrophic flow is 45° to the right (for $f > 0$) of the wind. This result depends on the form of the frictional parameterization, but not on the size of the viscosity.

At the edge of the Ekman layer the vertical velocity is given by (2.296), and so is proportional to the curl of the wind stress. (The second term on the right-hand side of (2.296) is the vertical velocity due to the divergence of the geostrophic flow, and is usually much smaller than the first term.) The production of a vertical velocity at the edge of the Ekman layer is one of most important effects of the layer, especially with regard to the large-scale circulation, for it provides an efficient means whereby surface fluxes are communicated to the interior flow (see Fig. 2.15).

2.12.5 Observations of the Ekman layer

Ekman layers — and in particular the Ekman spiral — are generally quite hard to observe, in either the ocean or atmosphere, both because of the presence of phenomena that are not included in the theory, and because of the technical difficulties of actually measuring the vector velocity profile, especially in the ocean. Ekman-layer theory does not take into account the effects of stratification or of inertial and gravity waves (section 2.10), nor does it account for the effects of convection or buoyancy-driven turbulence. If gravity waves are present, the instantaneous flow will be non-geostrophic and so time-averaging will be required to extract the geostrophic flow. If strong convection is present, the simple eddy-viscosity parameterizations used to derive the Ekman spiral will be rendered invalid, and the spiral Ekman profile cannot be expected to be observed in either atmosphere or ocean.

In the atmosphere, in convectively neutral cases, the Ekman profile can sometimes be qualitatively discerned. In convectively unstable situations the Ekman profile is generally not observed, but the flow is nevertheless cross-isobaric, from high pressure to low, consistent with the theory. (For most purposes, it is in any case the integral properties of the Ekman layer that is most important.) In the ocean, from about 1980 onwards improved instruments have made it possible to observe the vector current with depth, and to average that current

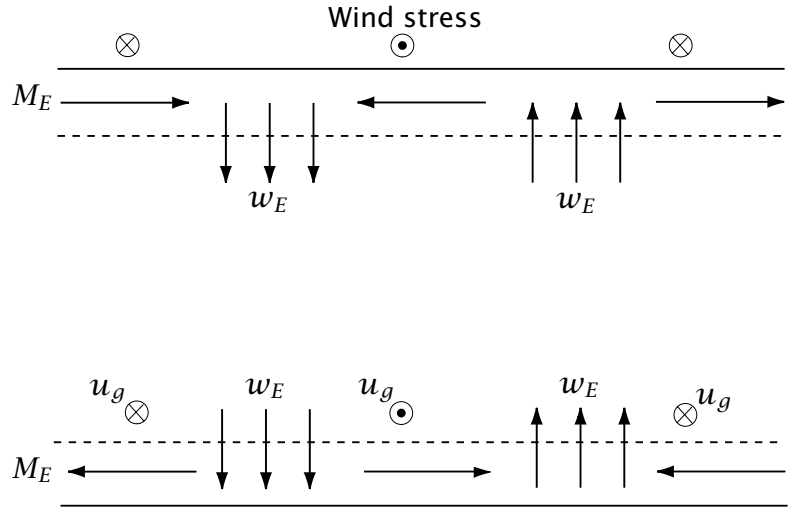


Fig. 2.15 Upper and lower Ekman layers. The upper Ekman layer in the ocean is primarily driven by an imposed wind stress, whereas the lower Ekman layer in the atmosphere or ocean largely results from the interaction of interior geostrophic velocity and a rigid lower surface. The upper part of figure shows the vertical Ekman 'pumping' velocities that result from the given wind stress, and the lower part of the figure shows the Ekman pumping velocities given the interior geostrophic flow.

and correlate it with the overlying wind, and a number of observations generally consistent with Ekman dynamics have emerged.[13] There are some differences between observations and theory, and these can be ascribed to the effects of stratification (which causes a shallowing and flattening of the spiral), and to the interaction of the Ekman spiral with turbulence (and the inadequacy of the eddy-diffusivity parameterization). In spite of these differences, Ekman layer theory remains a remarkable and enduring foundation of geophysical fluid dynamics.

2.12.6 * Frictional parameterization of the Ekman layer

[Some readers will be reading these sections on Ekman layers after having been introduced to quasi-geostrophic theory; this section is for them. Other readers may return to this section after reading chapter 5, or take (2.316) on faith.]

Suppose that the free atmosphere is described by the quasi-geostrophic vorticity equation,

$$\frac{\mathrm{D}\zeta_g}{\mathrm{D}t} = f_0 \frac{\partial w}{\partial z}, \tag{2.316}$$

where ζ_g is the geostrophic relative vorticity. Let us further model the atmosphere as a single homogeneous layer of thickness H lying above an Ekman layer of thickness $d \ll H$. If the vertical velocity is negligible at the top of the layer (at $z = H + d$) the equation of motion becomes

$$\frac{\mathrm{D}\zeta_g}{\mathrm{D}t} = \frac{f_0[w(H+d) - w(d)]}{H} = -\frac{f_0 d}{2H}\zeta_g \tag{2.317}$$

using (2.308). This equation shows that the Ekman layer acts as a *linear drag* on the interior flow, with a drag coefficient r equal to $f_0 d/2H$ and with associated time scale T_{Ek} given by

$$T_{Ek} = \frac{2H}{f_0 d} = \frac{2H}{\sqrt{2 f_0 A}}. \tag{2.318}$$

In the oceanic case the corresponding vorticity equation for the interior flow is

$$\frac{\mathrm{D}\zeta_g}{\mathrm{D}t} = \frac{1}{H}\mathrm{curl}_z \tau_s, \tag{2.319}$$

where τ_s is the surface stress. The surface stress thus acts as if it were a body force on the interior flow, and neither the Coriolis parameter nor the depth of the Ekman layer explicitly appear in this formula.

The Ekman layer is a very efficient way of communicating surface stresses to the interior. To see this, suppose that eddy mixing were the sole mechanism of transferring stress from the surface to the fluid interior, and there were no Ekman layer. Then the time scale of spindown of the fluid would be given by using

$$\frac{\mathrm{d}\zeta}{\mathrm{d}t} = A\frac{\partial^2 \zeta}{\partial z^2}, \tag{2.320}$$

implying a turbulent spin-down time, T_{turb}, of

$$T_{turb} \sim \frac{H^2}{A}, \tag{2.321}$$

where H is the depth over which we require a spin-down. This is much longer than the spin-down of a fluid that has an Ekman layer, for we have

$$\frac{T_{turb}}{T_{Ek}} = \frac{(H^2/A)}{(2H/f_0 d)} = \frac{H}{d} \gg 1, \tag{2.322}$$

using $d = \sqrt{2A/f_0}$. The effects of friction are enhanced because of the presence of a secondary circulation confined to the Ekman layers (as in Fig. 2.15) in which the vertical scales are much smaller than those in the fluid interior and so where viscous effects become significant; these frictional stresses are then communicated to the fluid interior via the induced vertical velocities at the edge of the Ekman layers.

Notes

1 The distinction between Coriolis force and acceleration is not always made in the literature. For a fluid in geostrophic balance, one might either say that there is a balance between the pressure force and the Coriolis force, with no net acceleration, or that the pressure force produces a Coriolis acceleration. The descriptions are equivalent, because of Newton's second law, but should not be conflated.

The Coriolis forces is named after Gaspard Gustave de Coriolis (1792–1843), who introduced the force in the context of rotating mechanical systems (Coriolis 1832, 1835). See Persson (1998) for a historical account and interpretation.

2 Phillips (1973). There is also a related discussion in Stommel & Moore (1989).

3 Phillips (1966). See White (2003) for a review. In the early days of numerical modelling these equations were the most primitive — i.e., the least filtered — equations that could practically be integrated numerically. Associated with increasing computer power there is a tendency for comprehensive numerical models to use non-hydrostatic equations of motion that do not make the shallow-fluid or traditional approximations, and it is conceivable that the meaning of the word 'primitive' may evolve to accommodate them.

4 The Boussinesq approximation is named for Boussinesq (1903), although similar approximations were used earlier by Oberbeck (1879, 1888). Spiegel & Veronis (1960) give a physically based derivation for an ideal gas, and Mihaljan (1962) provides a somewhat more general asymptotic derivation. Mahrt (1986) discusses applicability to the atmosphere.

5 As pointed out to me by W. R. Young.

6 Various versions of anelastic equations exist — see Batchelor (1953a), Ogura & Phillips (1962), Gough (1969), Gilman & Glatzmaier (1981), Lipps & Hemler (1982), and Durran (1989), although not all have potential vorticity and energy conservation laws (Bannon 1995, 1996; Scinocca & Shepherd 1992). The system we derive is most similar to that of Ogura & Phillips (1962) and unpublished notes by J. S. A. Green. The connection between the Boussinesq and anelastic equations is discussed by, among others, Lilly (1996) and Ingersoll (2005).

7 A numerical model that includes sound waves must take very small timesteps in order to maintain numerical stability, in particular to satisfy the Courant–Friedrichs–Lewy (CFL) criterion. An alternative is to use an implicit timestepping scheme that effectively lets the numerics do the filtering of the sound waves, and this approach is favoured by many numerical modellers. If we make the hydrostatic approximation then all sound waves except those that propagate horizontally are eliminated, and there is little need, vis-à-vis the numerics, to also make the anelastic approximation.

8 The Rossby number is named for C.-G. Rossby (see endnote on page 241), but it was also used by Kibel (1940) and is sometimes called the Kibel or Rossby-Kibel number. The notion of geostrophic balance and so, implicitly, that of a small Rossby number, predates either Rossby or Kibel.

9 After Taylor (1921b) and Proudman (1916). The Taylor–Proudman effect is sometimes called the Taylor–Proudman 'theorem', but it is more usefully thought of as a physical effect, with manifestations even when the conditions for its satisfaction are not precisely met.

10 Foster (1972).

11 Many numerical models of the large-scale circulation in the atmosphere and ocean do make the hydrostatic approximation. In these models convection must be *parameterized;* otherwise, it would simply occur at the smallest scale available, namely the size of the numerical grid, and this type of unphysical behaviour should be avoided. Of course in non-hydrostatic models convection must also be parameterized if the horizontal resolution of the model is too coarse to properly resolve the convective scales. See also problem 2.11.

12 After Ekman (1905). The problem was posed to V. W. Ekman (1874-1954), a student of Vilhelm Bjerknes, by Fridtjof Nansen, the polar explorer and statesman, who wanted to understand the motion of pack ice and of his ship, the *Fram*, embedded in the ice.

13 For oceanic observations see Davis *et al.* (1981), Price *et al.* (1987), Rudnick & Weller (1993). For the atmosphere see, e.g., Nicholls (1985).